Benton Tripp
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Work Experience

Quantitative Analyst – Tennessee Valley Authority
Chattanooga, TN (Remote)
August 2022 - Present

- Provide comprehensive analytics support, including developing three large-scale R Shiny applications to enhance Groundwater Evaluation, Geochemical Modeling, and Billing Anomaly Detection, significantly improving automation of advanced analytics across operations.
- Undertake proof-of-concepts on advanced analytics, such as topic modeling of Help Desk interactions and avian nesting patterns on transmission assets.
- Instruct two advanced R courses through the organization's Learning Management System.
- Develop tools for document parsing and analysis, integrating large language model (LLM) use cases and retrieval-augmented generation (RAG) systems.
- Utilize Azure DevOps, AWS cloud services, and the MLOps platform Dataiku to develop, deploy, and manage scalable machine learning pipelines for various projects.

Data Analyst – Elder Research Inc.
Raleigh, NC
August 2021 - August 2022

- Identified major slippage drivers for Church & Dwight, resulting in over $1 million in savings, while utilizing Databricks as part of the data pipeline for advanced analytics.
- Reconciled multiple inventory and sales forecasts for Chick-fil-A and established an end-to-end data pipeline using AWS Redshift, Athena, R, and Apache Airflow to streamline data processing.

Data Science Intern – Merkle Inc.
New York City, NY (Remote)
June 2021 - August 2021

- Created target audiences for marketing campaigns using advanced statistical and data modeling techniques.
- Conducted audience overlap analysis for the 2021 Jaguar/Land Rover vehicle campaigns.
- Assisted in calculating propensity scores for target audiences.
- Worked with Databricks for data analysis and pipeline management.

Forecasting Analyst – Genworth Financial
Lynchburg, VA
January 2020 - March 2021

- Analyzed call center data and developed a time-series forecasting model to predict future workload and staffing requirements.
- Developed models using Python and PostgreSQL, capturing trends, seasonality, and exogenous factors affecting call volume.
- Supported quantitative research efforts in response to the COVID-19 pandemic.

Education

Bachelor of Science in Data Analytics
Western Governor's University
Graduation: Fall, 2021

Master of Science in Geospatial Information Science and Technology *(In Progress)*
North Carolina State University
Expected Graduation: Spring, 2025

Certifications

Graduate Certificate in Applied Statistics and Data Management
North Carolina State University
Completed April 2024

AWS Cloud Practitioner
Amazon Web Services (AWS)
https://www.credly.com/badges/8308ddf1-24dd-482b-9535-0c73e3fde98f/
Issued July 2024

IT Information Library Foundations Certification (ITIL)
AXELOS Global Best Practice
Credential ID GR671287950BT
Issued Jun 2021

Independent Research

Species Distribution Modeling

- Conducted a study comparing machine learning models with traditional models in Species Distribution Modeling (SDM), such as Inhomogeneous Poisson Point Process and MaxEnt models.
- Used data from 8 bird species across 4 US states, focusing on bioclimatic variables.

Urban Growth Monitoring in Johnston County, NC

- Conducted a proof-of-concept study integrating Sentinel-2 multispectral satellite imagery with the Clay Foundation Model, an open-source deep learning framework, to monitor urban growth in Johnston County, NC.
- Designed a scalable and data-efficient framework to generate urban imperviousness estimates with greater temporal resolution than traditional datasets like the National Land Cover Database. Preprocessed Sentinel-2 imagery into structured geospatial datasets and extracted embeddings using the Clay Foundation Model.
- Developed and evaluated simple and deep neural network architectures, optimized through hyperparameter tuning, achieving high accuracy in urban imperviousness predictions.
- This methodology demonstrated potential for improving the frequency and responsiveness of urban monitoring, contributing to sustainable urban planning and resource management.

Skills

Programming Languages
Python, R, SAS, SQL/NoSQL, Javascript/HTML/CSS

Software/Technologies

- Git (Azure DevOps, GitHub)
- Cloud Technologies (Azure DevOps, AWS, Databricks)
- Dashboards (Power BI, Shiny, Dash)
- GIS Software (ArcGIS Pro, QGIS, GRASS GIS)
- MLOps Software (Dataiku, DataRobot)

Other Skills/Experience

- ETL and Data Integration
- Statistics & Applied Mathematics
- Machine Learning
- Forecasting
- Geospatial Analytics & Spatial Statistics